higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

ELECTRICAL PRINCIPLES AND PRACTICE
NQF LEVEL 4

(12041004)

27 February 2019 (Y-Paper)
13:00–16:00

This question paper consists of 7 pages and a formula sheet of 3 pages.

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TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Write neatly and legibly.

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QUESTION 1

1.1 Explain the following terms with reference to a sine wave:

1.1.1 Form factor

1.1.2 Peak factor

(2 × 2) (4)

1.2

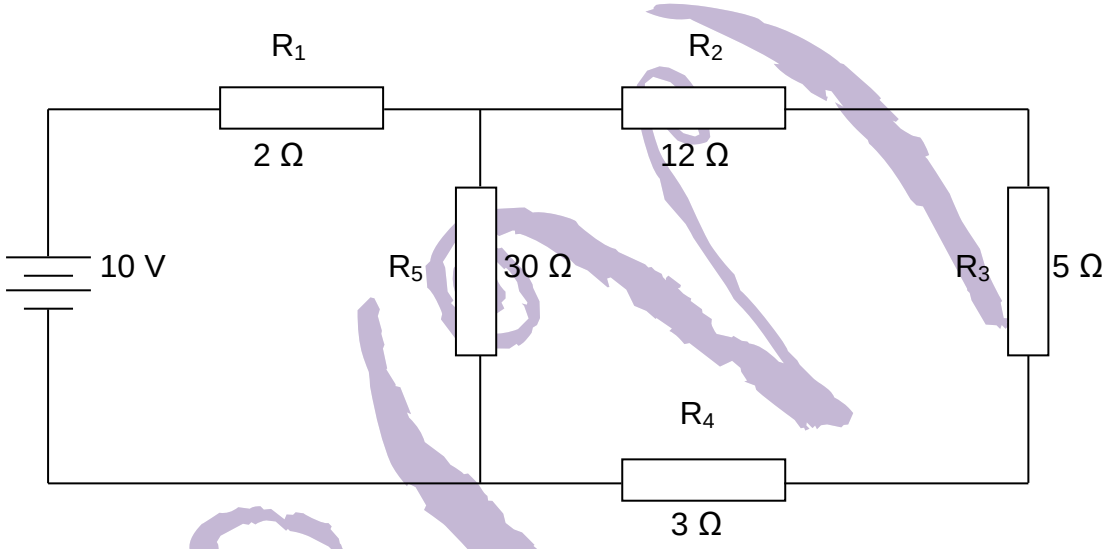


FIGURE 1

Refer to FIGURE 1 and calculate the following:

1.2.1 The total resistance of the circuit (5)

1.2.2 The total current flowing through the circuit (2)

1.2.3 The current flowing through the 30 Ω resistor (3)

1.2.4 The total resistance of the circuit if the resistor R_5 is a short circuit (1)

1.2.5 Explain your answer in Question 1.2.4. (2)

1.3 A single-phase machine draws 10 kVA at a power factor of 0,85 lagging.

Calculate the active power of the machine in kW. (3)

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1.4 A coil which has a resistance of 12 ohms and an inductance of 0,3 henry is connected in series with a capacitor of 30 microfarads. The circuit is connected to a 220VAC, 50Hz supply.

Calculate:

1.4.1 The capacitive reactance of the circuit (2)

1.4.2 The inductive reactance of the circuit (2)

1.5 There are two types of instrument transformers used in the electrical industry, namely, current transformers and voltage transformers.

State the function of an instrument transformer. (2)

1.6 Sir Charles Wheatstone was an English scientist responsible for many scientific breakthroughs in the Victorian era.

1.6.1 What is the Wheatstone bridge used for?

1.6.2 Name ONE common application of a Wheatstone bridge in the electrical industry.

(2 × 1) (2)

1.7 State how each of the following measuring instruments are connected in a low-voltage system:

1.7.1 A frequency meter

1.7.2 An ohmmeter

(2 × 1) (2)

[30]

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QUESTION 2

2.1 FIGURE 2 shows the layout of an electrical reticulation network (transmission and distribution of electrical power).

Name the sections indicated in the diagram by only writing the answer next to the letter (A–G) in the ANSWER BOOK. Do NOT redraw the sketch.

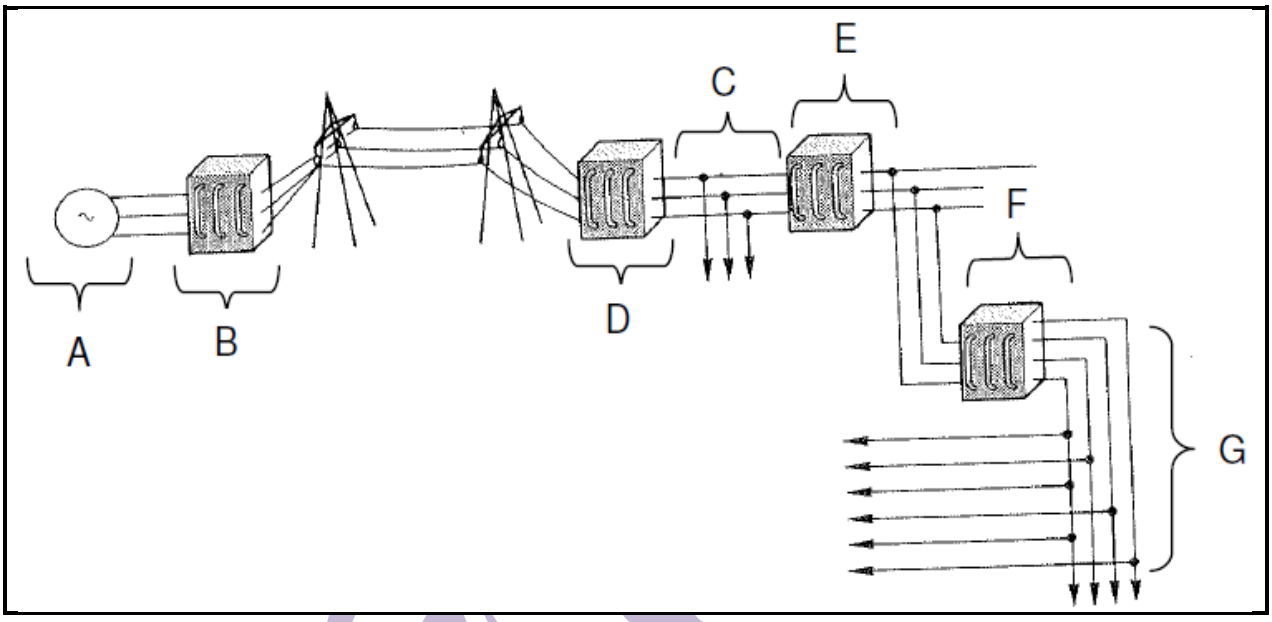


FIGURE 2

(7 × 1) (7)

2.2 Three-phase systems are connected in either star or delta.

Make a sketch of the phasor diagram of a balanced delta-connected load. (5)

2.3 Refer to the sketch in FIGURE 3 and answer the questions.



FIGURE 3

2.3.1 Name the component shown in FIGURE 3. (1)

2.3.2 Explain the function of the component you mentioned in QUESTION 2.3.1. (2)

[15]

QUESTION 3

3.1 Electrical devices and installations should be earthed according to SANS.

State the THREE functions of earthing. (3)

3.2 An earth electrode is one or more conductive parts that are embedded in the earth to make effective electrical contact with the general mass of the earth.

Name THREE materials that can be used in the manufacture of earth electrodes. (3)

3.3 3.3.1 Explain what you understand by *consumer's earth terminal.* (1)

3.3.2 Name TWO connection requirement of the consumer's earth terminal. (2)

3.4 Name FOUR parts that must be bonded in a domestic installation according to SANS 10142-1:2012. (4)

3.5 Name TWO earthing systems which are permitted for use in South Africa. (2)

[15]

QUESTION 4

4.1 Three single-phase transformers are connected to form a three-phase transformer.

Name FOUR common three-phase transformer configurations with which you are familiar. (4)

4.2 Calculate the efficiency at full load of a 150 kVA transformer operating at a power factor of 0,87, assuming the copper and iron losses to be 1 kW each. (6)

4.3 A three-phase transformer has a rating of 400 kVA and is connected in delta-star. The primary line voltage is 3,3 kV and the secondary line voltage is 380 V.

Calculate the following:

4.3.1 The turns ratio (5)

4.3.2 The primary line current (2)

4.3.3 The secondary line current (2)

4.3.4 The secondary phase current (1)

[20]

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QUESTION 5

5.1 Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'True' or 'False' next to the question number (5.1.1–5.1.3) in the ANSWER BOOK.

5.1.1 The multi-meter displays the peak to peak value of the AC voltage being measured.

5.1.2 Rotating magnetic fields are key to the operation of most AC motors.

5.1.3 A break in one phase affects the performance of induction motors.

(3 × 1) (3)

5.2 Name FOUR components of a split-phase motor. (4)

5.3 Use the table below to compare the characteristics of squirrel-cage and wound-rotor motors. Complete the table by only writing the answer next to the question number (5.3.1–5.3.6) in the ANSWER BOOK.

Characteristics	Squirrel-cage rotor motor	Wound-rotor motor
Starting torque	5.3.1	5.3.2
Starting current	5.3.3	5.3.4
Maintenance	5.3.5	5.3.6

(6 × 1) (6)

5.4 An overload protection device is used to safeguard a motor circuit against excessive current.

Name TWO types of overload protection devices. (2)

5.5 A DC series motor has a high starting and operating torque.

Make a neat sketch of the load characteristic curve of a series motor, clearly showing the relationship between speed and armature current. (3)

5.6 The function of a DC machine is to convert one form of energy into another.

Explain the basic difference between the following types of DC machines:

5.6.1 A generator

5.6.2 A motor

(2 × 1) (2)

[20]

TOTAL: 100

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FORMULA SHEET – ELECTRICAL PRINCIPLES AND PRACTICE

1. $\boldsymbol{Cos\phi = \frac{R}{Z}}$

2. $\boldsymbol{P = V\ I\ Cos\phi}$

3. $Q = V\ I\ Sin\phi$

4. $\boldsymbol{S = V\ I}$

5. $V_1 = \left(\frac{R_1}{R_t}\right) V_T$

6. $\boldsymbol{I_1 = \left(\frac{R_2}{R_1 + R_2}\right) I_T}$

7. $R_T = R_1 + R_2 + R_3 + \cdots + R_n$

8. $\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \cdots + \frac{1}{R_n}$

9. $\frac{V_1}{V_2} = \frac{N_1}{N_2} = \frac{I_2}{I_1}$

10. $V_L = \sqrt{3}\, V_{ph}$

11. $\boldsymbol{E = V - I_a R_a}$

12. $n_s = \frac{f}{p}$

13. $S = \frac{N_s - N_r}{N_s}$

14. $\boldsymbol{R_{sh} = \frac{I_m R_m}{I_{sh}}}$

15. $R_{se} = \frac{V_t}{I_t} - R_m$

16 $\tan\boldsymbol{\phi} = \sqrt{3}\left(\frac{\boldsymbol{W_1 - W_2}}{\boldsymbol{W_1 + W_2}}\right)$

17 $E = V + I\,r$

18 $\boldsymbol{V = I\,R}$

19 $R = \frac{\rho\, l}{A}$

20 $\boldsymbol{R_t = R_0(1 + \alpha t)}$

21 $R_t = R_0[1 + \propto_0 (t - \emptyset)]$

22 $P = V\ I$

23 $E = I^2\ R\,t$

24 $\boldsymbol{\eta = \frac{P_0}{P_{in}} \times 100}$

25 $\boldsymbol{X_L = 2\pi\, f\, L}$

26 $X_C = \frac{1}{2\pi\, f\, C}$

27 $\boldsymbol{Z = \sqrt{R^2 + (X_L - X_C)^2}}$

28 $Mmf = N\ I$

29 $H = \frac{Mmf}{l}$

30 $B = \frac{\Phi}{A}$

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31 $F = B\,I\,l$

32 $E = -L\dfrac{di}{dt}$

33 $A = \dfrac{\pi\, d^2}{4}$

34 $E = \dfrac{1}{2}\ C\,V^2$

35 $Q = C\,V$

36 $F = 2\times10^{-7}\times\dfrac{I_1\ I_2}{d}$

37 $I = \dfrac{Emf}{R + r_T}$

38 $L = N\dfrac{\Delta\,\phi}{\Delta\,I}$

39 $v = V_m \sin(\omega t \pm \emptyset)$

40 $Z = \dfrac{V}{I}$

41 $I_T = \sqrt{I_R^2 + (I_L - I_C)^2}$

42 $tan\emptyset = \dfrac{X_L - X_C}{R}$

43 $I_{sh} = \dfrac{V + I_{se}R_{se}}{R_{sh}}$

44 $T = \dfrac{0{,}318 I_a Zp\emptyset}{c}$

45 $E = V - I_a(R_a + R_{se})$

46 $E = B\,l\,v$

47 $E = -N\ \dfrac{d\emptyset}{dt}$

48 $E = \dfrac{V}{d}$

49 $E = \dfrac{1}{2}LI^2$

50 $Q = It$

51 $C_T = C_1 + C_2 + C_3 + \cdots + C_n$

52 $\dfrac{1}{C_T} = \dfrac{1}{C_1} + \dfrac{1}{C_2} + \dfrac{1}{C_3} + \cdots + \dfrac{1}{C_n}$

53 $L_T = L_1 + L_2 + L_3 + \cdots + L_n$

54 $\dfrac{1}{L_T} = \dfrac{1}{L_1} + \dfrac{1}{L_2} + \dfrac{1}{L_3} + \cdots + \dfrac{1}{L_n}$

55 $tan\emptyset = \dfrac{I_L - I_C}{I_R}$

56 $E = V + I_a R_a$

57 $V = E - I(R_a + R_{se})$

58 $E = V + I_a(R_a + R_{se})$

59 $E = V + I_a R_a + IR_{se}$

60 $E = V - I_a R_a - IR_{se}$

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61 $$E = \frac{2p\emptyset ZN}{60\ c}$$

62 $$V_T = \sqrt{V_R^2 + (V_L - V_C)^2}$$

63 $$V_L = IX_L$$

64 $$E_{rms}\ or\ I_{rms} = 0{,}707\ E_m\ or\ I_m$$

65 $$N = \frac{V}{k\emptyset}$$

66 $$E = 4{,}44\ f\emptyset_m N$$

67 $$V_C = IX_C$$

68 $$E_{ave} or\ I_{ave} = 0{,}637\ E_m\ or\ I_m$$

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